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Lab Assignment #04

Aim:
Design and simulate a Magic Tee junction, investigate the effects of design parameter variations on its scattering parameters.

Theory:
The Magic Tee is a four-port microwave network that functions as a hybrid power divider and combiner. It is constructed by combining an E-plane Tee and an H-plane Tee into a single junction. The structure comprises a main waveguide section with two collinear arms (Ports 1 and 2), intersected by a shunt side arm (Port 3, H-arm) and a series side arm (Port 4, E-arm).

In this simulation, the waveguide is defined as Standard WR-137 using Air as the dielectric medium ($\epsilon r \approx 1$). The behavior of the junction is defined by the interaction between the dominant TE10 mode and the geometric symmetry of the intersection:

1. H-Arm (Sum Port - Port 3): When power enters the H-arm, it splits equally into the collinear arms (Ports 1 and 2) with the electric fields in phase (0° difference). This behavior mimics an H-plane Tee, where S13=S23.
2. E-Arm (Difference Port - Port 4): When power enters the E-arm, it splits equally into the collinear arms but with a 180° phase shift (E-field vectors are opposite). This mimics an E-plane Tee, where S14=−S24.
3. Isolation: A fundamental characteristic of the Magic Tee is the isolation between ports. Ideally, the E-arm and H-arm are decoupled (S34=0), and the two collinear ports are isolated from each other (S12=0) due to the symmetry of the mode structures.

$$S = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}$$

Fig 1.1 S matrix of Magic Tee Waveguide

Parametric Sweep and Modal Analysis
The performance of the Magic Tee, specifically its isolation and phase balance, is critically dependent on the broad wall dimension (a) of the WR-137 waveguide. By performing a parametric sweep with values of 25 mm, 34.85 mm (Standard WR-137), and 45 mm, we observe how physical geometry impacts wave propagation in an air-filled guide.

The cutoff frequency (fc) for the dominant TE10 mode in air is governed by fc=2ac.

- Reduced Width (25 mm): Reducing a to 25 mm shifts the cutoff frequency up to approximately 6 GHz. Since WR-137 is typically operated in the C-band (starting ~5.85 GHz), this constriction risks pushing the waveguide into an evanescent state for lower frequencies in the band. In a Magic Tee, this leads to high insertion loss and destroys the magnitude balance between the collinear arms.

- Standard Width (34.85 mm): At the standard WR-137 width of 34.85 mm, the cutoff frequency is 4.3 GHz. This provides a healthy safety margin for C-band operation, ensuring pure single-mode propagation. This dimension is optimal for the Magic Tee; it maintains the correct guide wavelength (λg) to support the constructive and destructive interference patterns required for the "Sum" and "Difference" operations.
- Increased Width (45 mm): Increasing the width to 45 mm lowers the cutoff to 3.33 GHz. While this lowers impedance, it allows the potential propagation of higher-order modes (such as TE20 or TE01) at C-band frequencies. These modes do not adhere to the strict symmetry required for the Magic Tee's isolation, causing signal leakage between the E and H ports (S34=0) and degrading the "magic" behavior.

Additionally, the characteristic impedance (Z0) is a function of the dimension a. At the standard 34.85 mm, the junction is physically optimized to minimize reflections (S11,S22,S33,S44). Deviating to 25 mm or 45 mm creates a significant impedance step at the junction, increasing return loss and introducing phase errors that deviate from the ideal 0° and 180° outputs.

Table 1 - Parametric Details of WR-137

Parameter	**Value**
Broad Dimension (Width)	34.8488 mm
Narrow Dimension (Height)	15.7988 mm
Cutoff Frequency of Lowest Order Mode:	4.301 GHz
Cutoff Frequency of Upper Mode:	8.603 GHz
Recommended Frequency Band:	5.85 to 8.20 GHz
Length	100 mm (suggested)
Wall Material	PEC
Core Material	Air

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Procedure:

Part 1: Modeling and Setup

1. Open CST Studio Suite and select the New Template option from the main dashboard.
2. Navigate to the Microwave & RF / Optical folder, choose Waveguides & Components, and then select Waveguides.
3. Choose the Time Domain Solver from the available options to handle the broadband frequency sweep.
4. Set the project units by defining Dimensions in mm and Frequency in GHz.
5. Set the simulation frequency range with a Minimum of 2 GHz and a Maximum of 8 GHz. This ensures you capture the TE10 cutoff and the C-band operation.
6. Open the Modeling tab and set the Background Material to PEC (Perfect Electric Conductor) with zero spacing in all directions.
7. Create parameters in the Parameter List for standard WR-137 dimensions:
    - Wide dimension: a=34.8488
    - Narrow dimension: b=15.7988
    - Length: l=100
8. Use the Brick tool to create the main horizontal waveguide section (Ports 1 & 2) using parameters a,b,and l. Assign the material as Air.
9. Create a second Brick for the E-plane side arm (Port 4), orienting it vertically on the broad wall of the main guide.
10. Create a third Brick for the H-plane side arm (Port 3), orienting it horizontally on the narrow wall of the main guide.
11. Merge all three bricks using the Boolean Add operation to create a single, continuous four-port junction.
12. Use the Pick Face tool to select the openings of the four arms and define them as Waveguide Ports 1, 2, 3, and 4.
13. Navigate to the Simulation tab and define Field Monitors for Electric (E) and Magnetic (H) fields at 6 GHz.
14. Open the Parametric Sweep tool, add parameter a as the sweep variable, and define the values: 25 mm, 34.8488 mm, and 45 mm.
15. Start the Setup Solver to run the simulation across all defined parametric steps.

Part 2: Post-Processing (Excitation Analysis)

16. Navigate to the Post-Processing tab and select Combine Results to simulate simultaneous port excitation.
17. Case 1: Sum Mode (H-Arm Output)

- Select the E-Field Monitor (6 GHz).
- Set Port 1 to Amplitude: 5, Phase: 0.
- Set Port 2 to Amplitude: 5, Phase: 0.
- Click Combine. Observe that the fields add up constructively and exit through Port 3 (H-Arm).

18. Case 2: Difference Mode (E-Arm Output)
    - Select the E-Field Monitor (6 GHz) again.
    - Set Port 1 to Amplitude: 5, Phase: 0.
    - Set Port 2 to Amplitude: 5, Phase: 180.
    - Click Combine. Observe that the fields interfere destructively at the center and exit through Port 4 (E-Arm).

Part 3: Results Visualization

19. Navigate to the 1D Results folder to view S-parameter plots (S11,S21,S31,S41) and verify the cutoff frequency shifts.
20. Navigate to 2D/3D Results to view the vector field distributions for the single-port excitations at 6 GHz.

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<u>3D Model:</u>

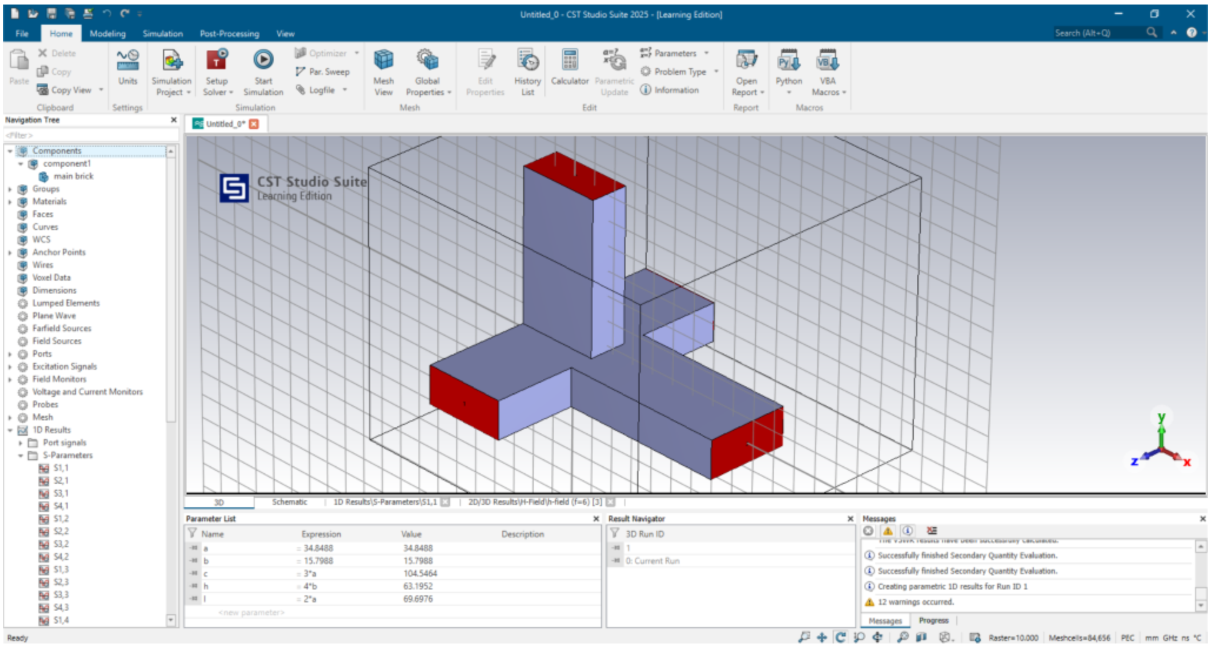

Fig 2.1 Magic Tee Junction 3D Model

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Results:

The electromagnetic characteristics of the WR-137 Magic Tee junction were simulated over a broadband frequency sweep from 2 GHz to 8 GHz. This bandwidth allows for the observation of the dominant TE10 mode cutoff as well as the operational bandwidth where port isolation (S34) and power splitting balance are optimized. Field distributions were captured specifically at 6 GHz to visualize the stable, propagating Sum and Difference modes within the standard C-band operation of the device.

Magic Tee Junction

Cutoff Frequency

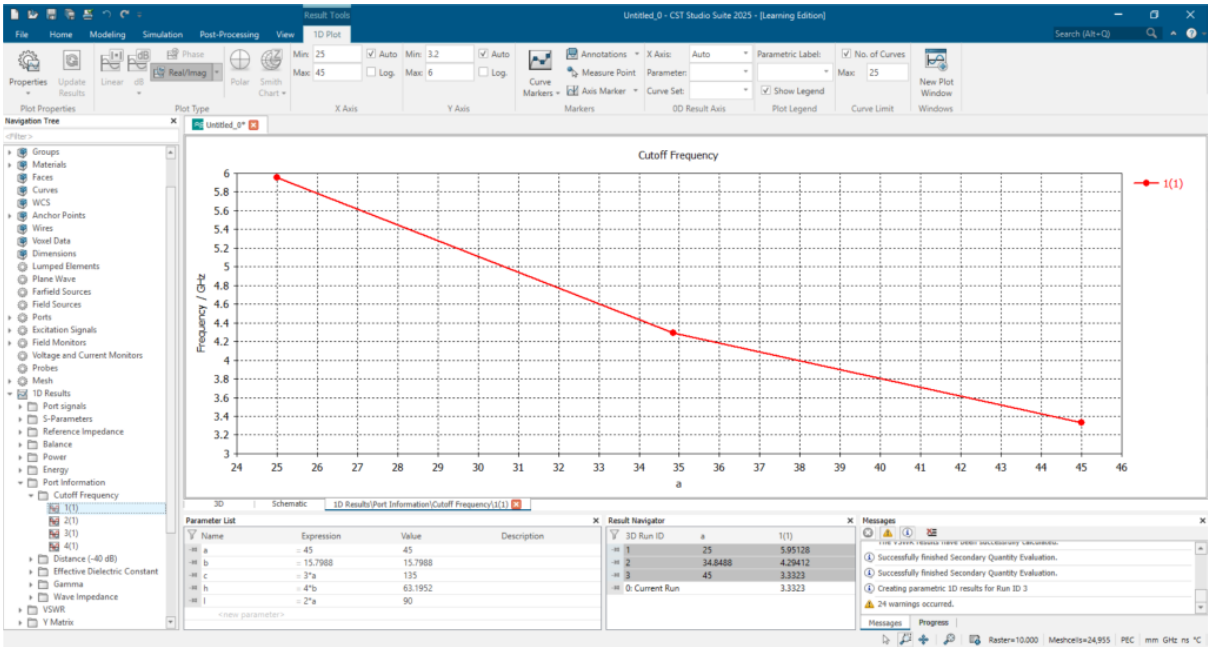

Fig 3.1 Observed cutoff frequencies for wider dimensions (a = 25, 34.8488 and 45 mm)

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Scattering Parameters

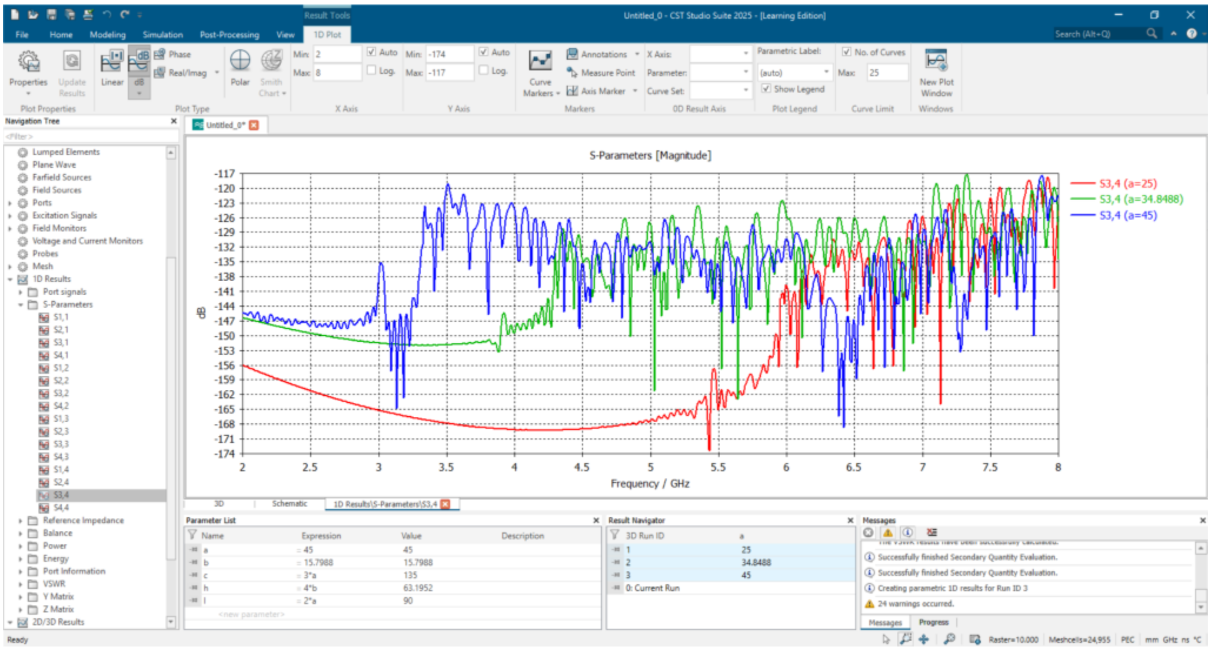

Fig 4.1 Scattering Parameter S(3,4) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

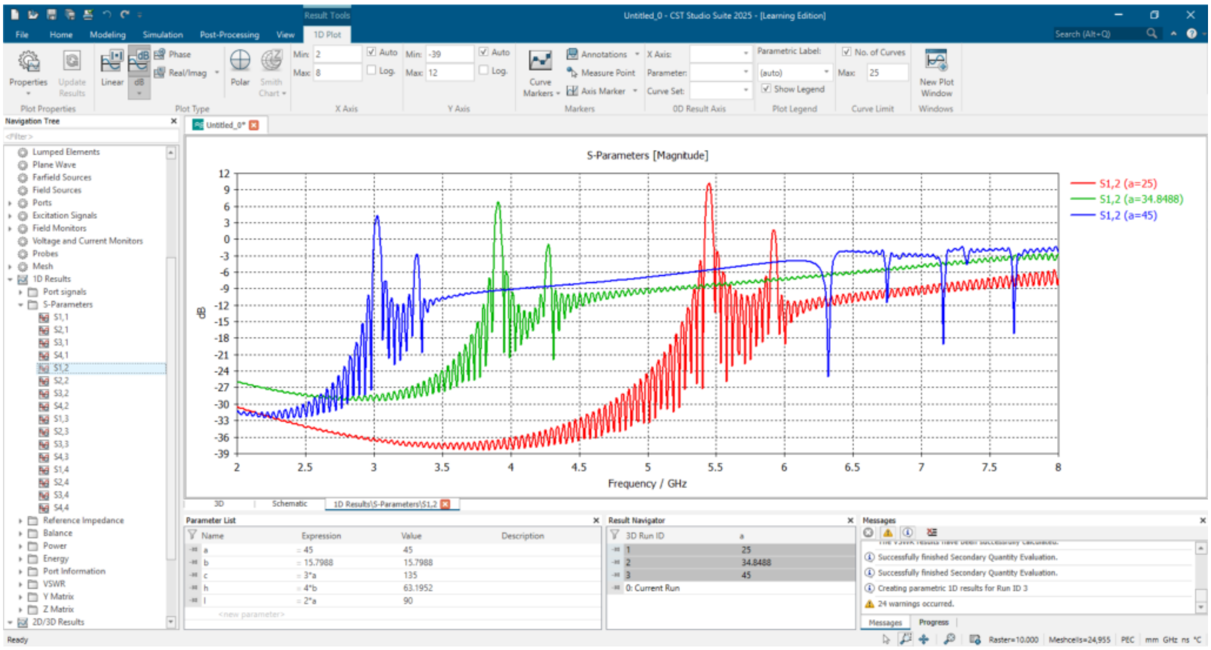

Fig 4.2 Scattering Parameter S(1,2) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

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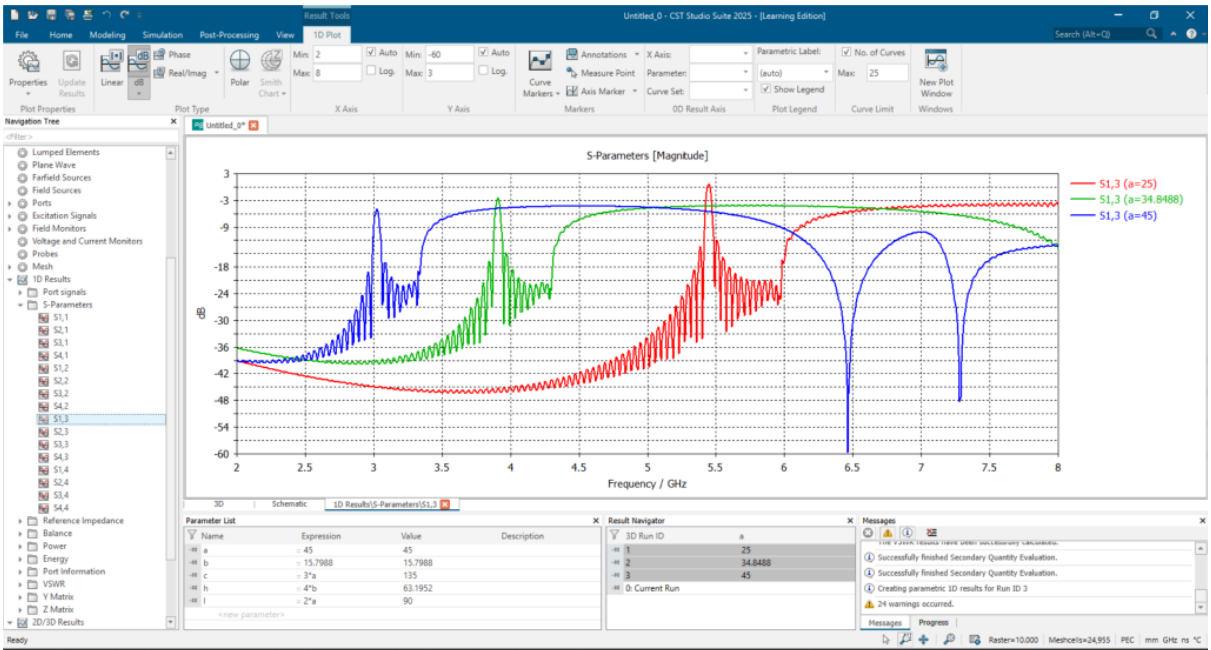

Fig 4.3 Scattering Parameter S(1,3) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

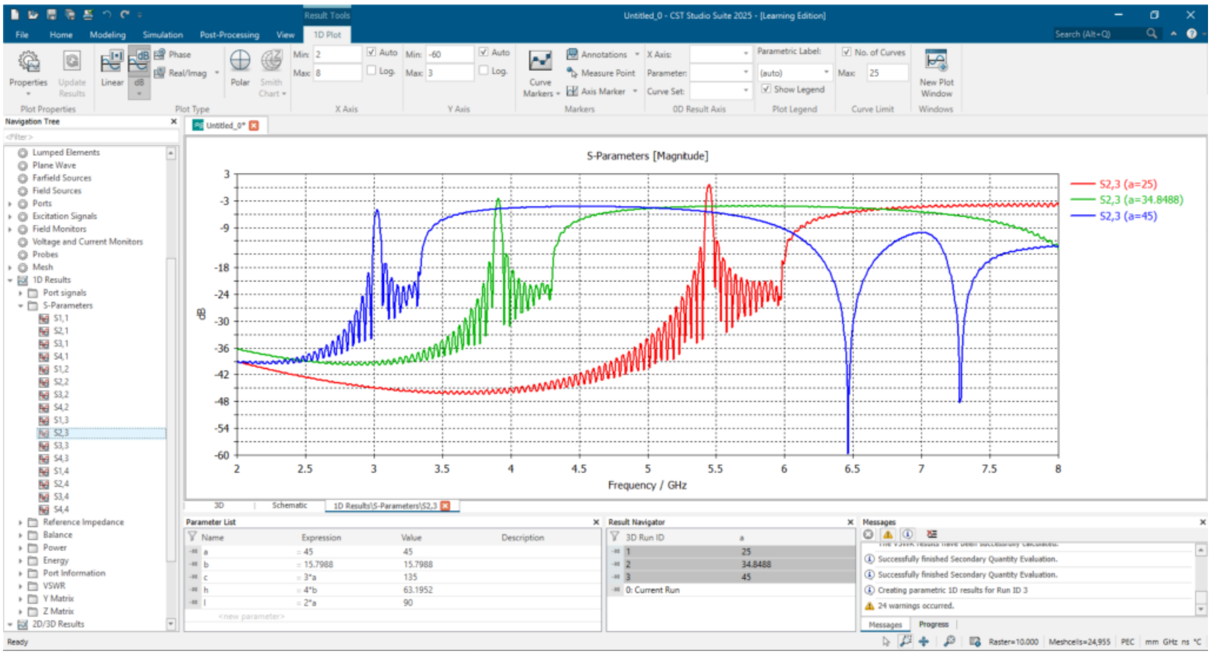

Fig 4.4 Scattering Parameter S(2,3) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

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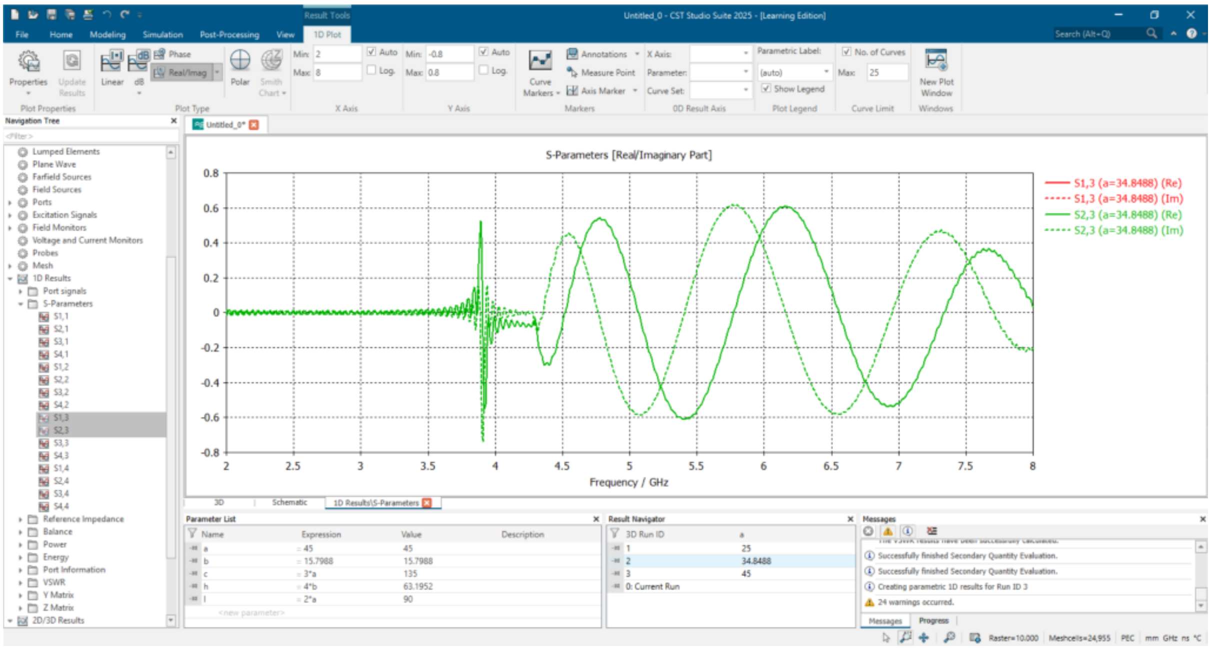

Fig 4.5 Zero phase difference observed between Scattering Parameter S(1,3) and S(2,3)

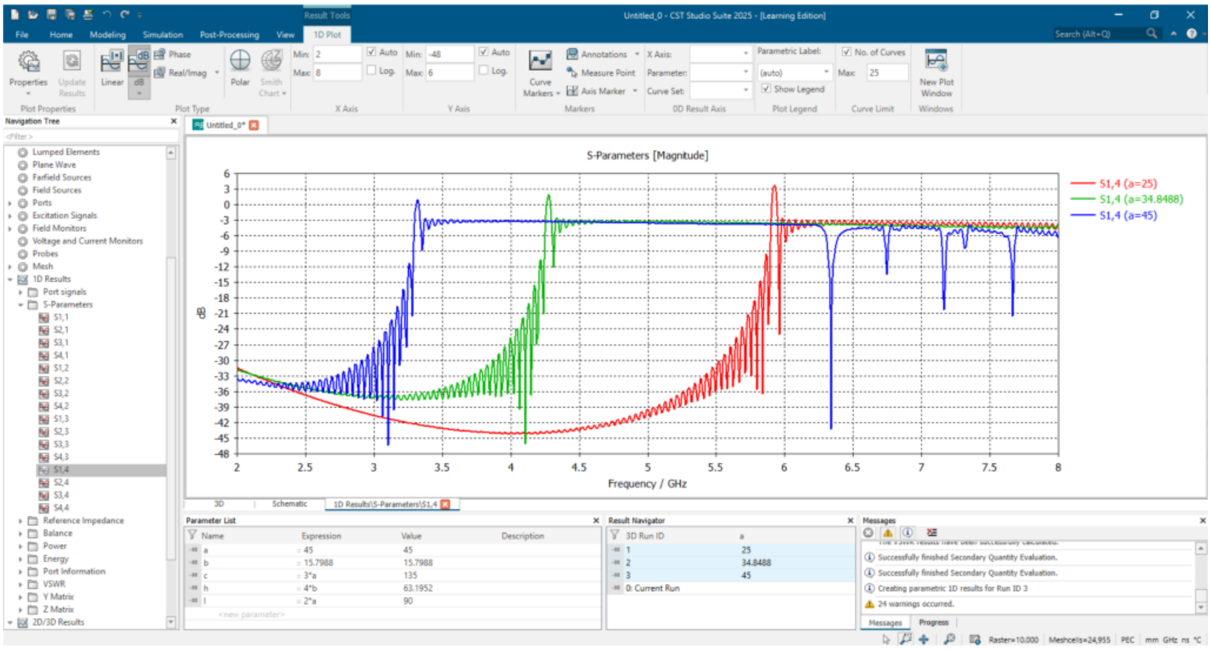

Fig 4.6 Scattering Parameter S(1,4) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

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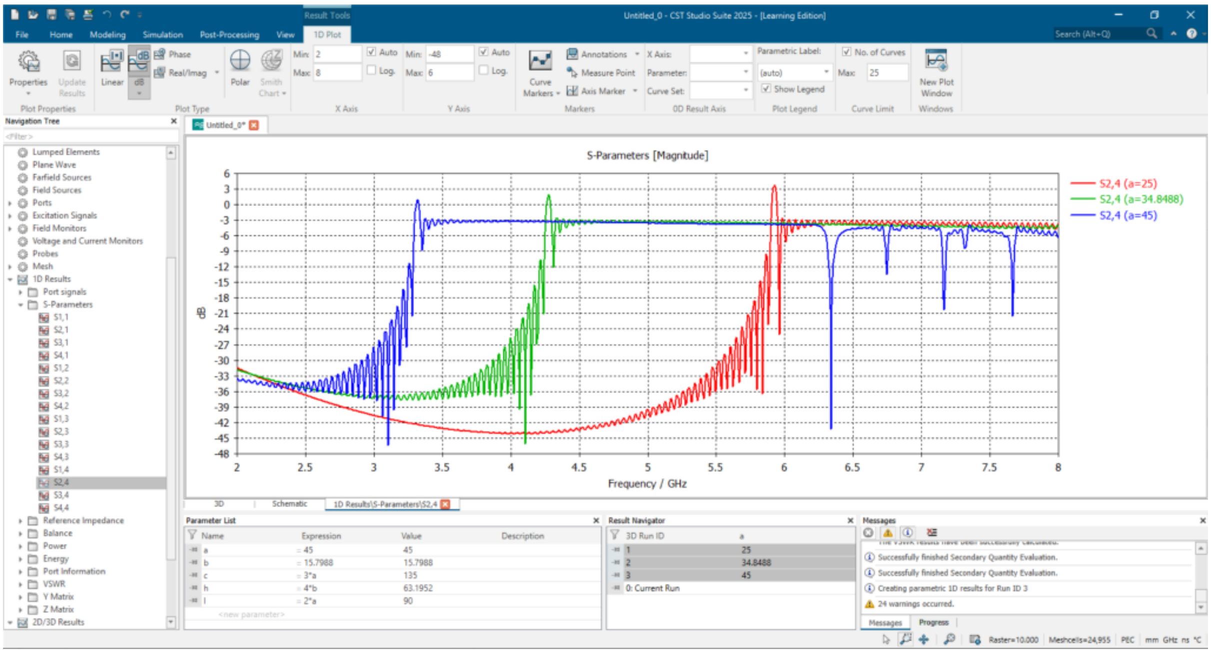

Fig 4.7 Scattering Parameter S(2,4) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

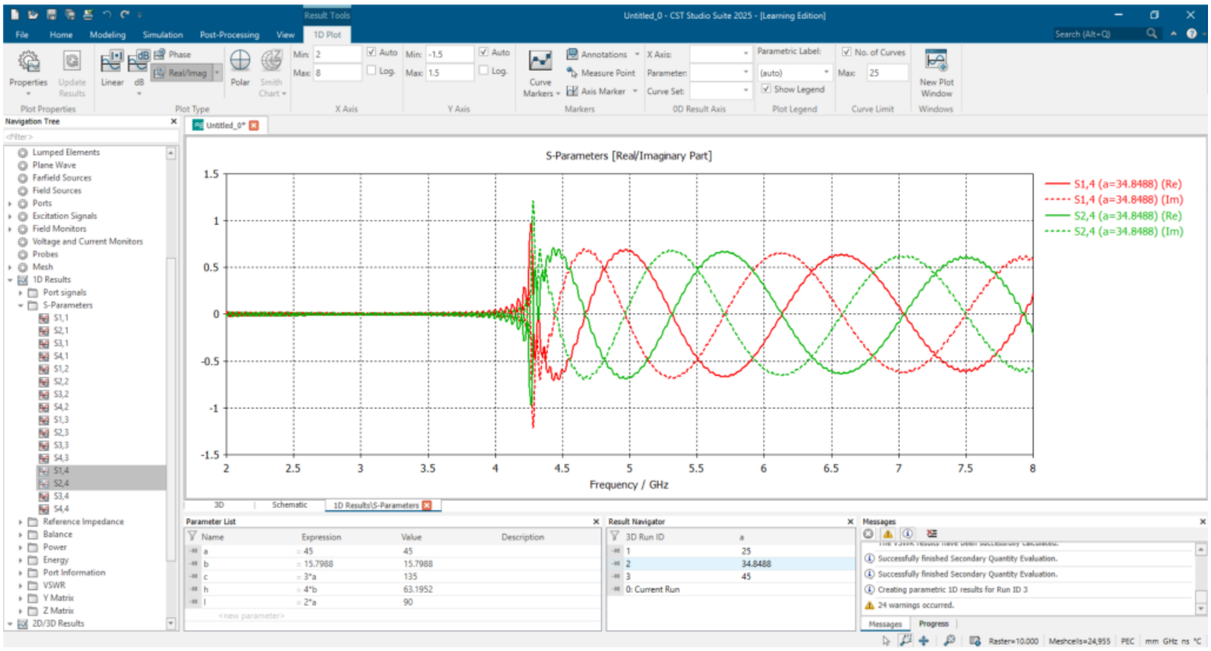

Fig 4.8 degree phase difference observed between Scattering Parameter S(1,4) and S(2,4)

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Electric Field Distributions

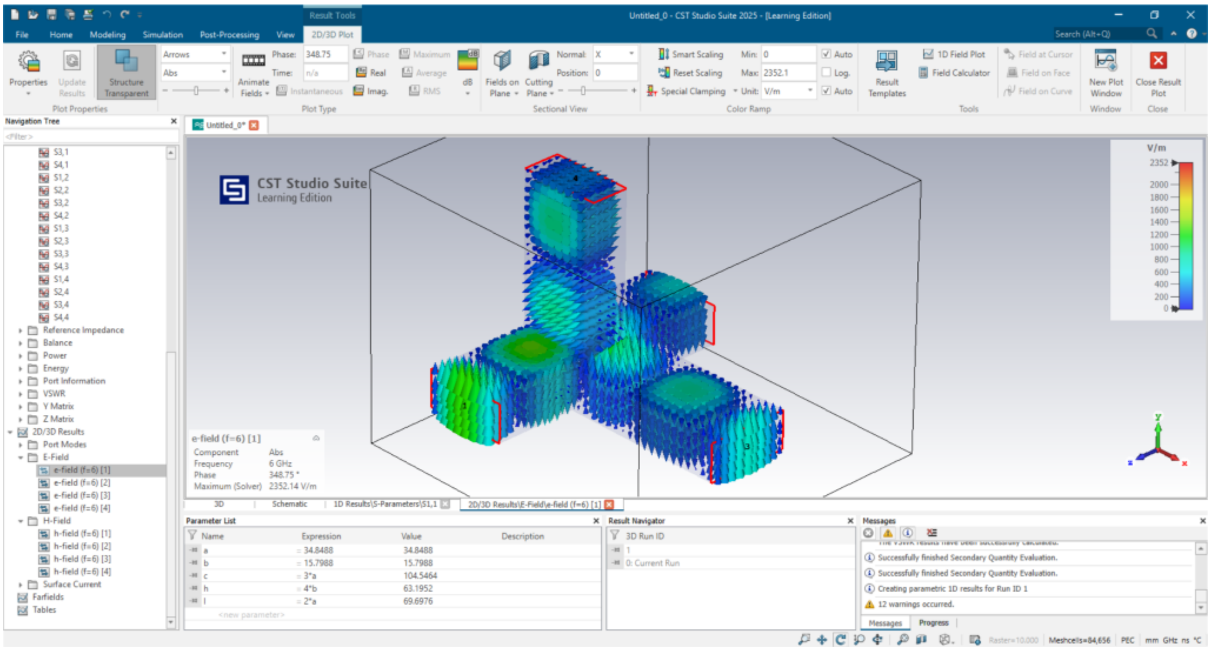

Fig 5.1 Electric field distribution for f = 6 GHz at port 1

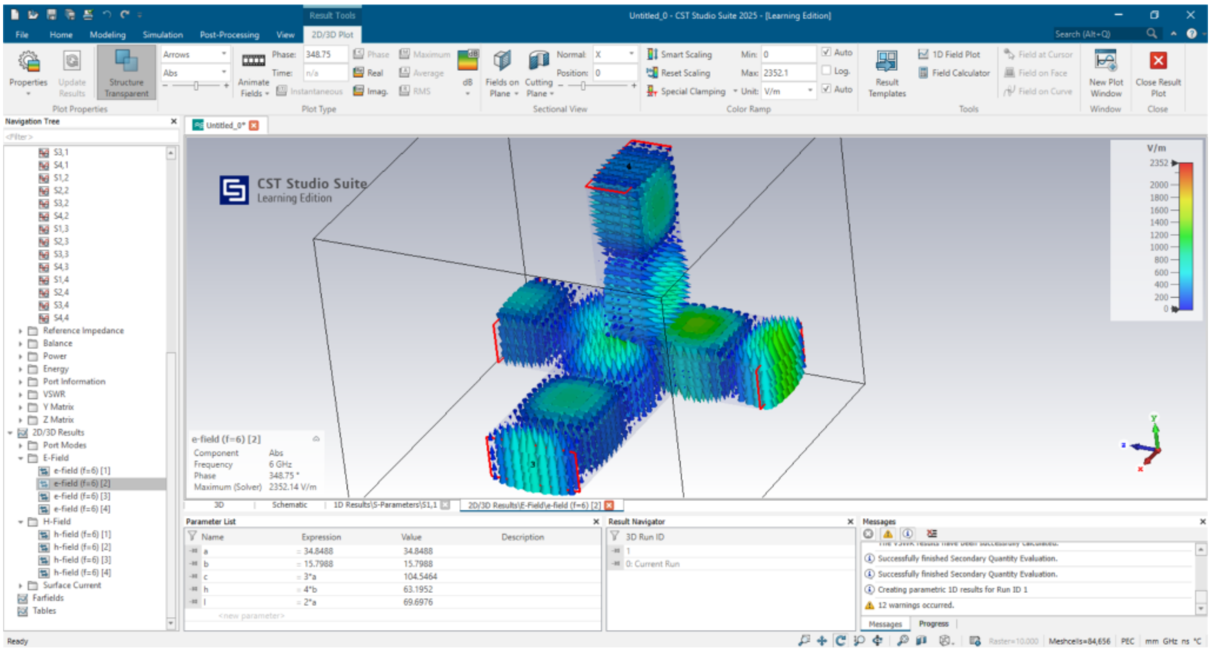

Fig 5.2 Electric field distribution for f = 6 GHz at port 2

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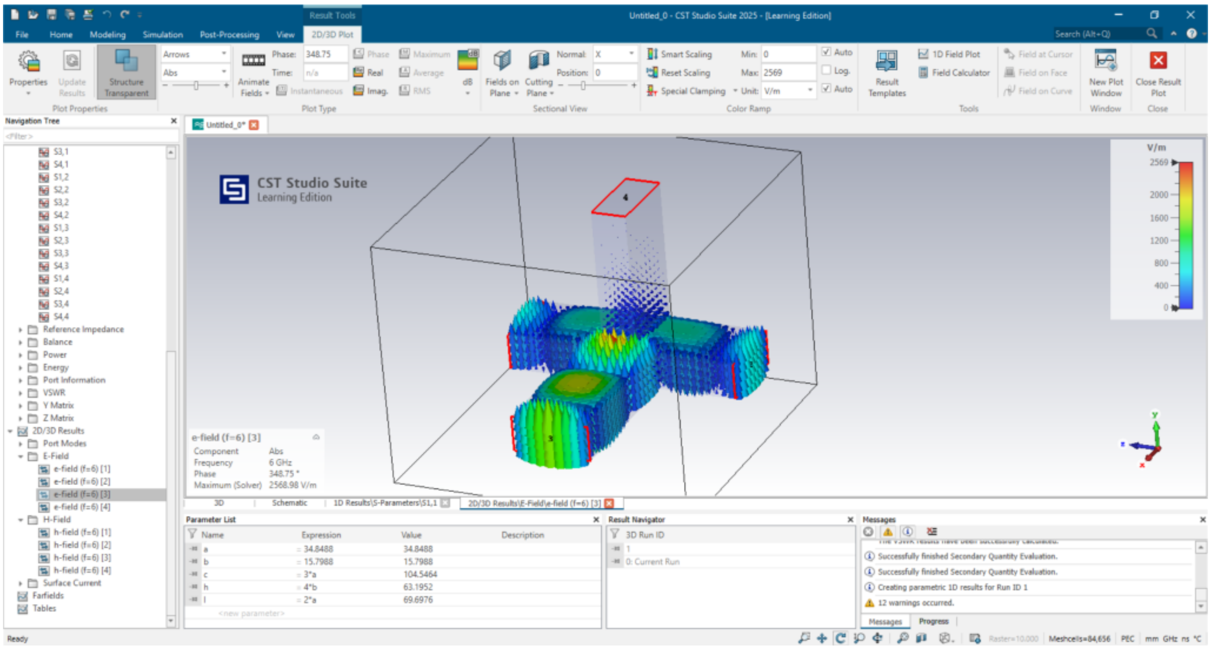

Fig 5.3 Electric field distribution for f = 6 GHz at port 3

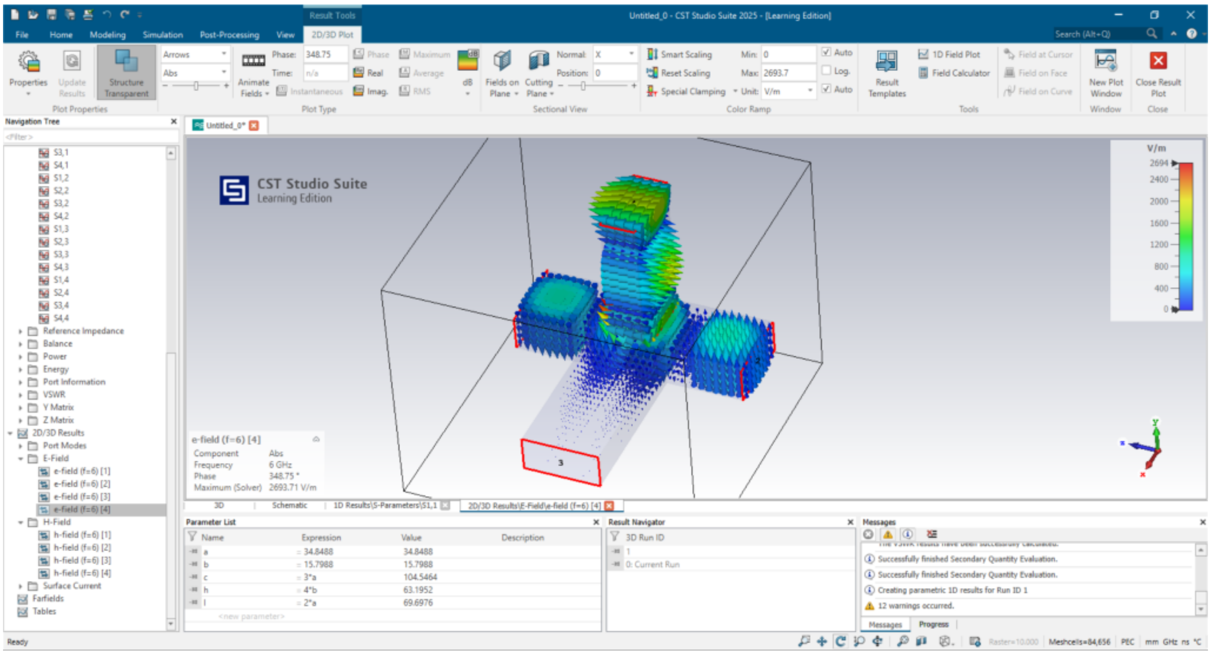

Fig 5.4 Electric field distribution for f = 6 GHz at port 4

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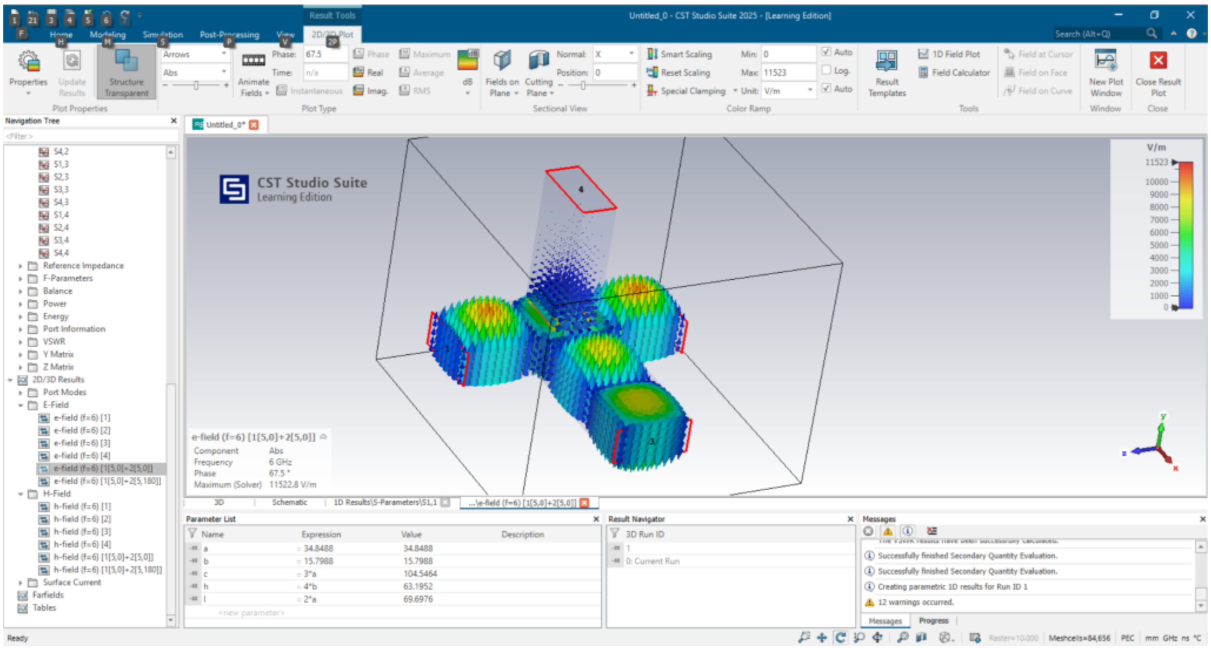

Fig 5.5 Electric field distribution for f = 6 GHz, when port 1 and 2 are excited with 5V amplitude and 0 degree phase shift

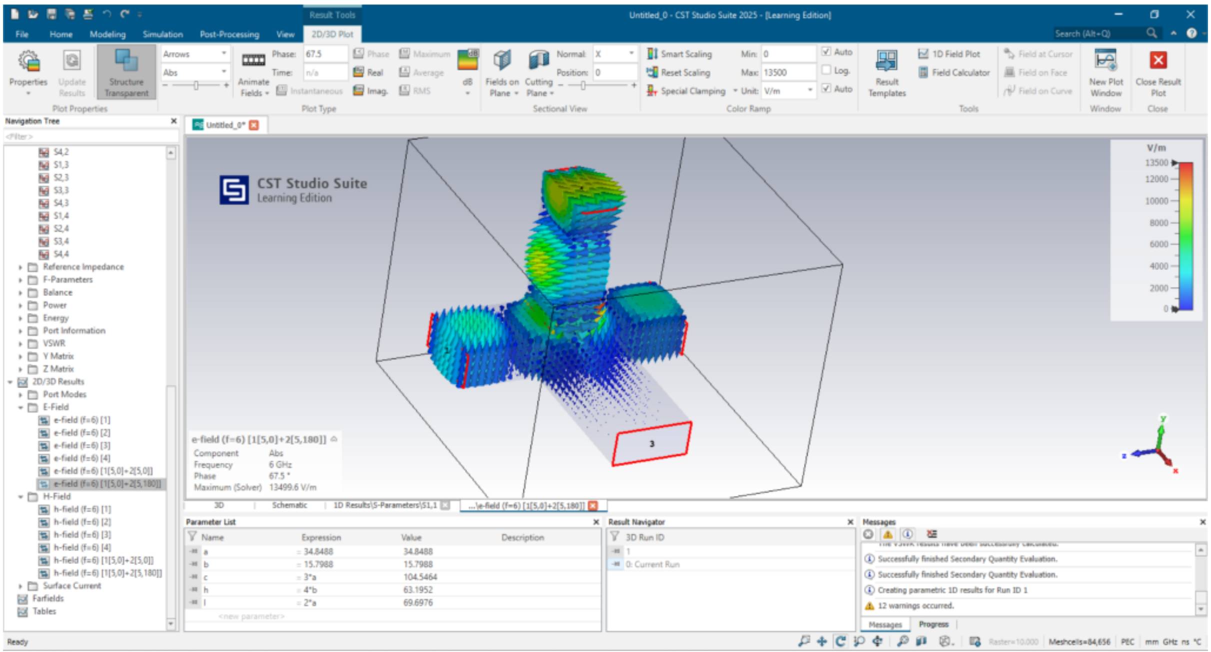

Fig 5.6 Electric field distribution for f = 6 GHz, when port 1 and 2 are excited with 5V amplitude and 180 degree phase shift

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Magnetic Field Distributions

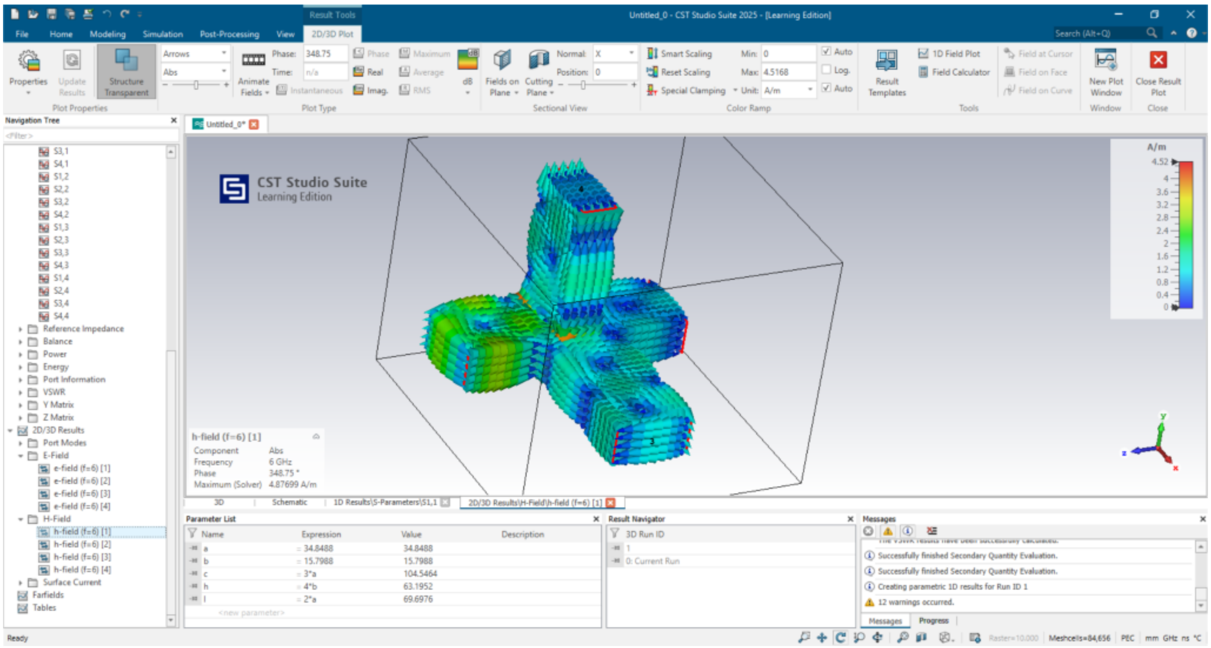

Fig 6.1 Magnetic field distribution for f = 6 GHz at port 1

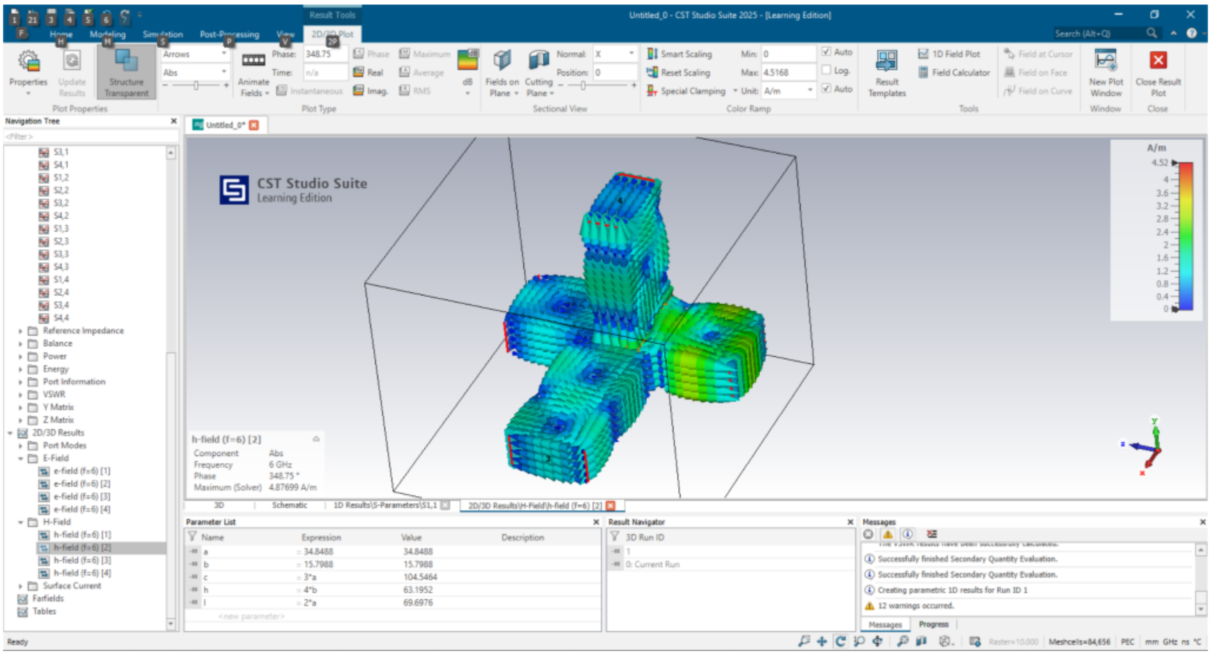

Fig 6.2 Magnetic field distribution for f = 6 GHz at port 2

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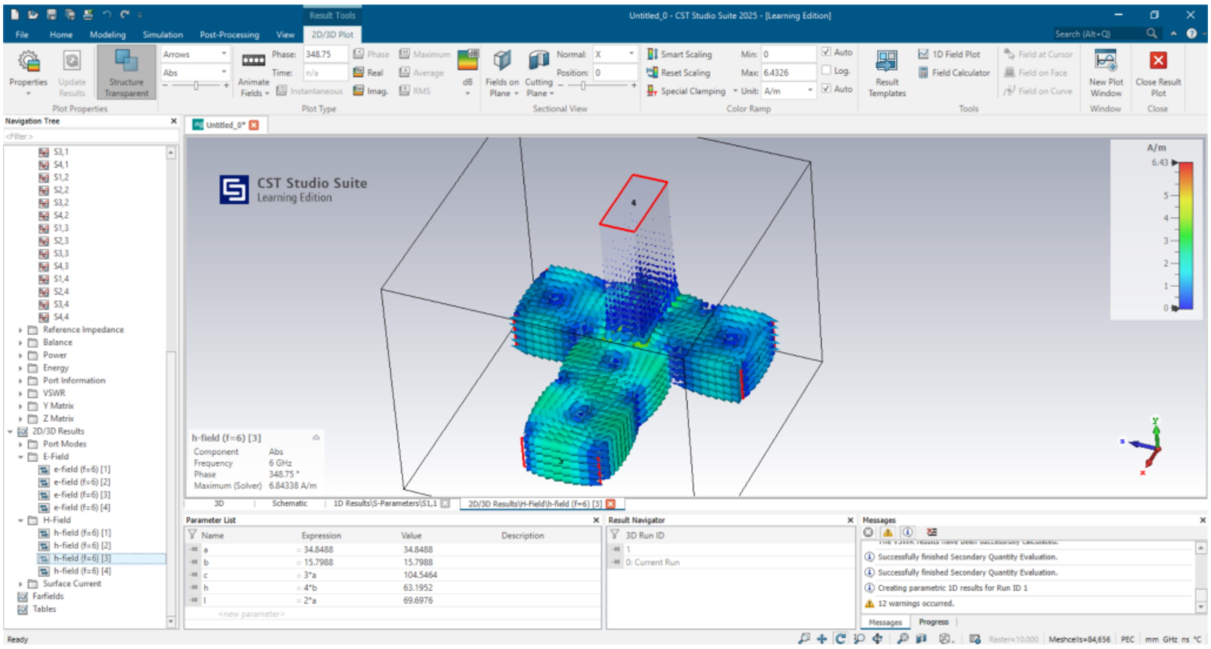

Fig 6.3 Magnetic field distribution for f = 6 GHz at port 3

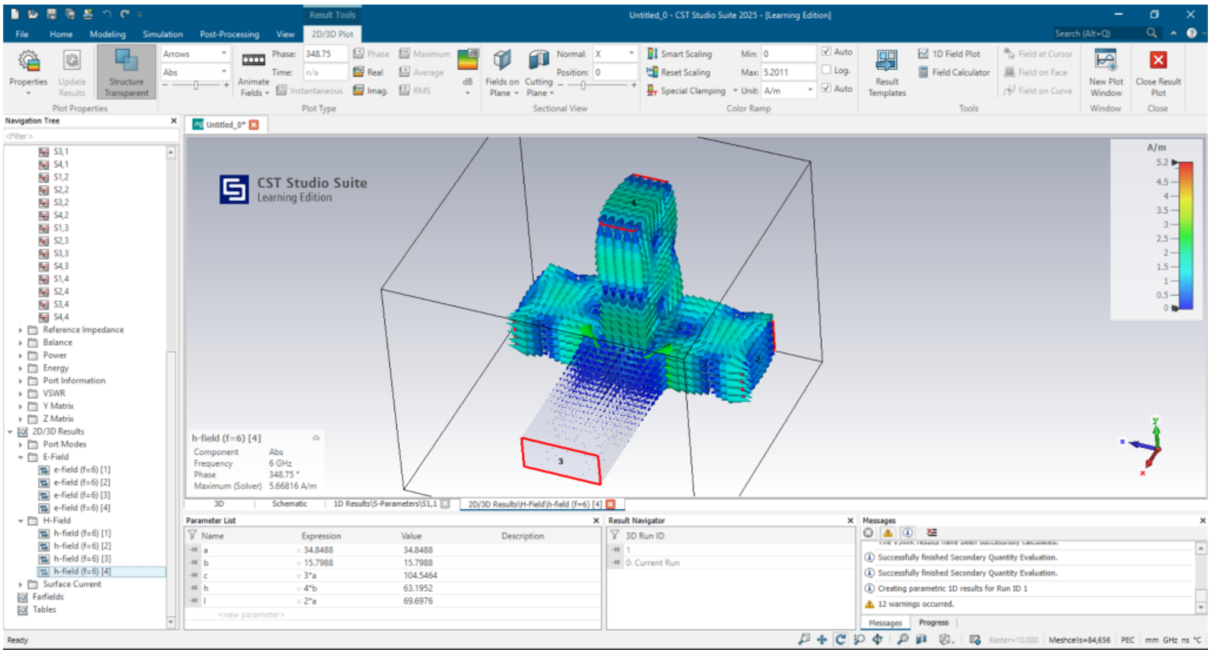

Fig 6.4 Magnetic field distribution for f = 6 GHz at port 4

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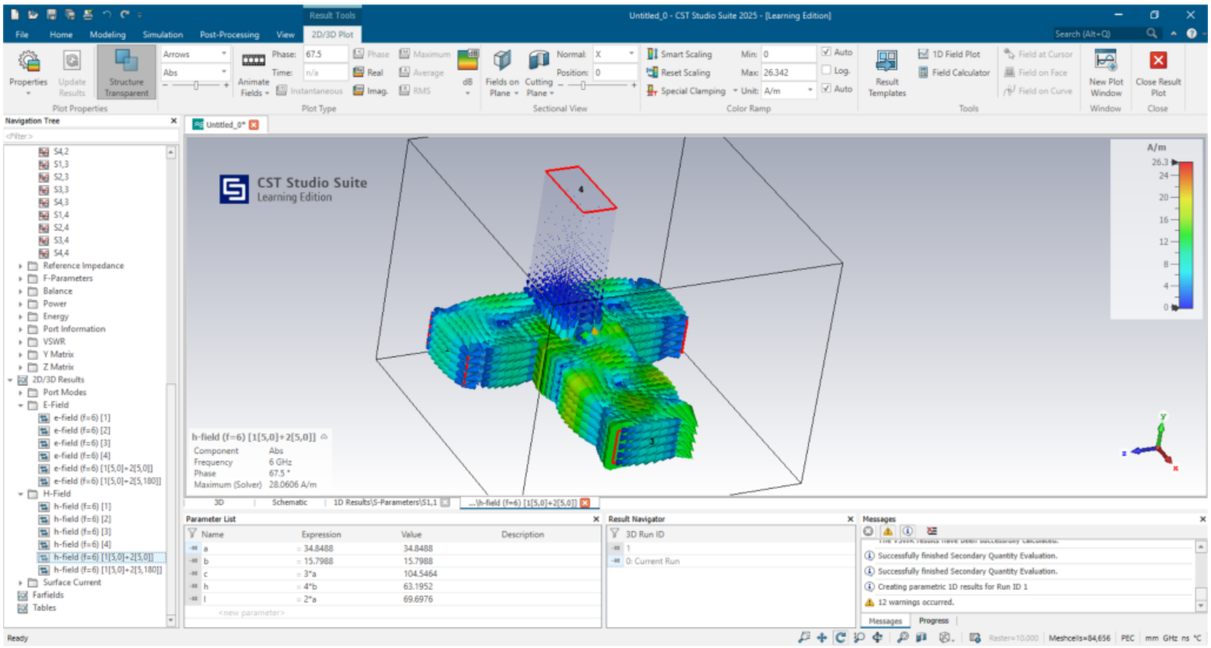

Fig 6.5 Magnetic field distribution for f = 6 GHz, when port 1 and 2 are excited with 5V amplitude and 0 degree phase shift

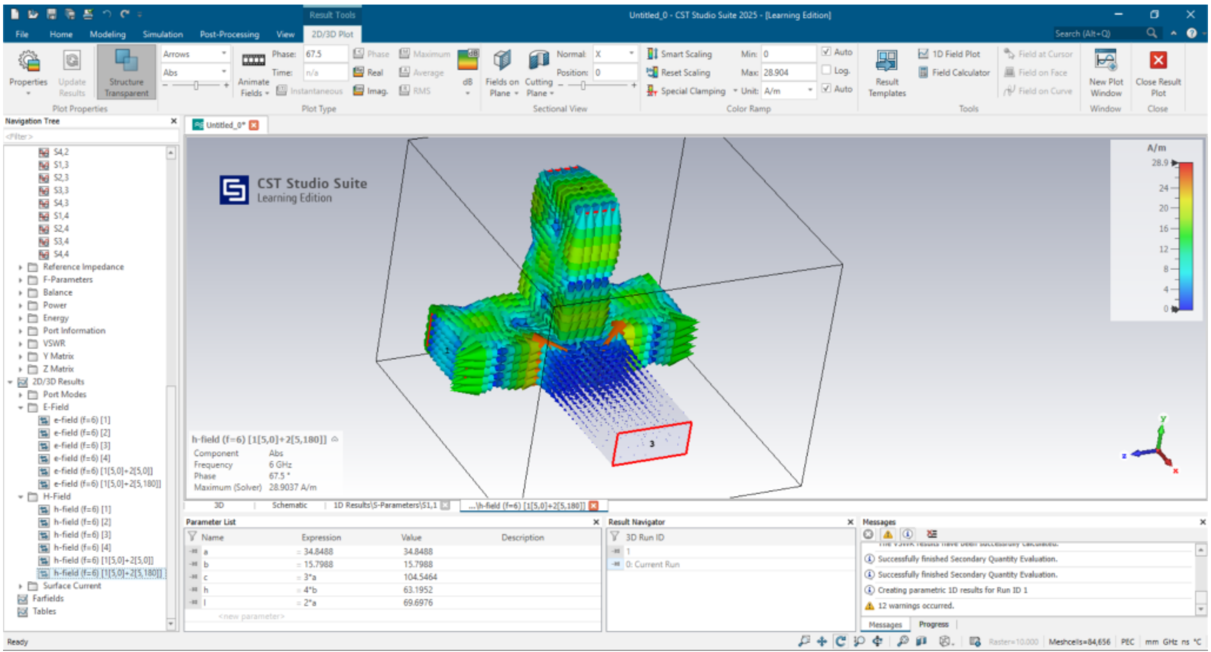

Fig 6.6 Magnetic field distribution for f = 6 GHz, when port 1 and 2 are excited with 5V amplitude and 180 degree phase shift

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Observations

S-Parameters and Cutoff Frequency Variations

The broadband frequency sweep from 2 GHz to 8 GHz provided critical insight into the relationship between the waveguide's physical width (a) and its propagation characteristics. As observed in the Cutoff Frequency plot (Fig 3.1) and the S-parameter magnitude plots, the "turn-on" frequency shifts significantly with the dimension a.

- For the 25 mm width (red curve): The cutoff frequency shifted upward to approximately 6 GHz, which is evident in Fig 3.1. Consequently, the S-parameter plots show that the signal is heavily attenuated below this mark, confirming that the waveguide is evanescent in the lower C-band.
- For the standard 34.85 mm width (green curve): The cutoff remained at the theoretical 4.3 GHz, ensuring stable operation across the recommended frequency band. This dimension demonstrated the optimal balance for coupling and isolation.
- For the 45 mm width (blue curve): The cutoff dropped to approximately 3.3 GHz. While this allows lower frequency propagation, the S-parameter plots reveal ripples and degraded isolation, suggestive of potential higher-order mode interference that disrupts the symmetry required for the Magic Tee's isolation.

Isolation and Phase Balance

The "Magic" properties of the junction were verified through the isolation and phase difference plots.

- Isolation (S34): Fig 4.1 illustrates the isolation between the Sum (Port 3) and Difference (Port 4) ports. The standard dimension (green curve) maintains high isolation (low magnitude in dB) across the operating band, whereas the 25 mm and 45 mm variations show significantly reduced performance.
- Phase Characteristics: The unique phase behavior was confirmed in Fig 4.5 and Fig 4.8. Fig 4.5 explicitly demonstrates a zero phase difference between the collinear arms when fed from the H-arm, confirming its in-phase nature. Conversely, Fig 4.8 confirms a 180-degree phase difference between the collinear arms when fed from the E-arm, validating its anti-symmetric behavior.

Sum Mode (H-Plane Behavior)

The H-arm's function as a coherent adder was validated through field distributions at 6 GHz. When Ports 1 and 2 were excited simultaneously with an amplitude of 5V and a 0-degree phase shift (in-phase), the constructive interference resulted in the strong field propagation emerging from Port 3, as shown in Fig 5.5. This visualizes the device's ability to sum the input signals and direct them to the H-plane port.

Difference Mode (E-Plane Behavior)

The E-arm's function as a subtractor was similarly confirmed. When Ports 1 and 2 were excited simultaneously with an amplitude of 5V but with a 180-degree phase shift (anti-phase), the fields interfered destructively at the center junction. This forced the energy to propagate solely out of Port

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4, as clearly observed in the Electric field distribution in Fig 5.6. This confirms the device's ability to output the difference between the two input signals.

Conclusions:

The simulation of the Magic Tee junction using standard WR-137 waveguide dimensions was successfully validated over the 2 GHz to 8 GHz frequency range. The analysis of scattering parameters and field distributions confirmed that the device functions correctly as a hybrid coupler, satisfying the fundamental conditions of isolation, power splitting, and phase balance within the C-band.

The simulation demonstrated the dual nature of the Magic Tee. The H-plane arm (Port 3) functioned effectively as a "Sum" port, producing in-phase outputs at the collinear arms. Conversely, the E-plane arm (Port 4) operated as a "Difference" port, generating the required 180-degree phase shift. This behavior was visually corroborated by the post-processing field combinations (Fig 5.5 and Fig 5.6), which showed distinct constructive interference at the Sum port and destructive interference directing energy to the Difference port.

The parametric sweep revealed that performance is highly sensitive to the broad wall dimension (a). The standard width of 34.85 mm was proven optimal, providing a cutoff frequency of 4.3 GHz that supports the dominant TE10 mode while suppressing higher-order modes. Deviations to 25 mm pushed the cutoff too high (6 GHz), rendering the device evanescent in the lower band. Increasing the width to 45 mm introduced mode instability and degraded isolation.

Overall, the standard WR-137 Magic Tee design exhibited excellent isolation between the E and H arms (S34) and minimal return loss at the design frequency. The agreement between the theoretical scattering matrix and the simulated results verifies the device's suitability for applications requiring precise signal combining and difference extraction.